

NandhaKumar.P

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ABOUT ME

I'm an undergraduate Computer Science student with a strong interest in Machine Learning and Deep Learning. I have basic knowledge of HTML, CSS and JavaScript, and hands-on experience with TensorFlow for building and training models and OpenCV for image processing. I'm passionate about applying AI techniques to solve real-world problems and continuously exploring new technologies.

EDUCATION

Amrita School of Computing	2022 – 2026
B.Tech in Computer Science & Engineering	
Vivekam Matric Higher Secondary School	2022
SRMV Higher Secondary School	2020

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

Machine Learning & AI: Scikit-Learn, TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transfer Learning, Computer Vision

GCP: Cloud Shell, Cloud Shell Editor, Compute Engine (VM Instances), Machine Images, Managed Instance Groups (MIG), App Engine

AWS: EC2, S3, IAM, VPC, Lambda, RDS, DynamoDB, SES, AMI

Tools & Platforms: OpenCV, Git, GitHub, Jupyter Notebook, Google Colab

Databases: MongoDB

EXPERIENCE

Machine Learning Engineer / Researcher — Data Conquest Research Hub, Coimbatore Jan 2026 – May 2026

- Worked on end-to-end Machine Learning and Deep Learning solutions involving data preprocessing, feature engineering, model development, hyperparameter tuning, and performance optimization for real-world AI applications.
- Designed and implemented deep learning architectures using TensorFlow, PyTorch, and OpenCV for computer vision, multimodal learning, time-series analysis, and predictive analytics.
- Developed scalable data processing pipelines including data cleaning, augmentation, normalization, dataset preparation, and automated training workflows for large-scale image and sensor datasets.
- Evaluated and optimized machine learning models using performance metrics such as Accuracy, Precision, Recall, F1-Score, Dice Coefficient, IoU, ROC-AUC, PR-AUC, RMSE, MAE, and R^2 .
- Performed literature review, model benchmarking, ablation studies, experimental analysis, and technical documentation to support research-driven AI development and reproducible experimentation.
- Collaborated on the development of production-ready AI solutions by integrating model inference pipelines, explainable AI techniques, and deployment-oriented workflows for real-world applications.

PROJECTS

Retrieval-Grounded Multimodal Clinical Language Modeling 2025

Developed a privacy-preserving multimodal healthcare AI framework integrating Retrieval-Augmented Generation (RAG), patient-specific vector indexing, structured clinical records, and diagnostic imaging for evidence-grounded clinical language modeling. Designed an offline pipeline with modality-aware visual feature extraction and local LLM-based generation to improve factual consistency, multimodal reasoning, and data privacy.

AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems 2026

Designed a lightweight deep learning framework for Remaining Useful Life prediction and health-state classification using multivariate sensor data from NASA N-CMAPSS. Integrated regime-aware sensor gating, Graph Attention Networks, Mamba-based temporal modeling, and multitask learning to improve predictive performance while enabling efficient edge-IIoT deployment.

FitVerse (Health & Fitness Mobile Application) 2024

Developed a cross-platform mobile application using React Native, Firebase, and Android Studio for health and fitness monitoring. Implemented step tracking, hydration and sleep reminders, activity monitoring, health analytics, and chatbot assistance with Google Fit integration for real-time health data synchronization.

Driver Drowsiness Detection using Computer Vision 2024

Built a real-time driver monitoring system using OpenCV and facial landmark detection to identify driver fatigue and inattention from live video streams. Implemented image preprocessing, eye-state analysis, and alert generation mechanisms, and evaluated the system using accuracy, precision, recall, and F1-score.

RESEARCH WORK

A Secure Deep Learning Framework for Image and Video Deepfake Detection with Immutable Result Storage July 2025 - March 2026

Developing a secure deepfake detection framework combining multiple deep learning architectures and a tamper-resistant deployment pipeline. The system evaluates three models: a hybrid XceptionNet + Vision Transformer for combining local texture analysis with global attention, VGG16 enhanced with frequency-domain features to capture compression artifacts, and DenseNet to improve multi-scale feature sharing through dense connectivity. A custom dataset is generated using InsightFace-based face swapping, where source and target identities are merged to create synthetic deepfake samples that are then used for training and benchmarking. The project also implements a real-time secure web application using Streamlit with MongoDB storage to ensure immutable detection results and practical deployment. The framework performs systematic model comparison, integrates spatial and frequency features, and focuses on building a reliable, real-world deepfake detection system.

CERTIFICATIONS

Tata Group Data Analytics Job Simulation – Forage

Cloud Computing Foundations – Google Cloud

Deep Learning for Developers – Infosys Springboard